Henry Qi

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Education

University of California, Berkeley Berkeley, CA
*B.S. in Electrical Engineering and Computer Science (EECS) — **GPA: 3.75/4.00*** *Aug. 2022 – May 2026*

- **Relevant Coursework:** Machine Learning (CS 189), Efficient Algorithms (CS 170), Discrete Math & Probability (CS 70), Probability & Random Processes (EECS 126), Operating Systems (CS 162), Database Systems (CS 186)

Experience

Intel Corporation Folsom, CA
Software & Automation Engineering Intern *Aug. 2026 – Dec. 2026*

- Conceptualized and built a domain-specific RAG pipeline indexing technical chip specifications and command protocols to power an autonomous validation AI agent, automating hardware triage workflows.
- Engineered a CLI developer automation tool to eliminate repetitive command lookups, reducing onboarding toil and execution friction across engineering teams.
- Integrated structured logging and error handling into internal automation workflows, improving hardware debugging speed and system test visibility.

Copatible Berkeley, CA
Lead Backend Engineer *Jan. 2026 – May 2026*

- Engineered a high-throughput Java backend pipeline, optimizing data processing workflows and improving system throughput and stream efficiency by 30% under production-scale workloads.
- Implemented secure multi-tenant data isolation architecture, enhancing data protection and reducing cross-tenant risk by 40% while accelerating regression testing workflows by 25%.

UC Berkeley Computer Science Department Berkeley, CA
Computer Science Tutor *Jan. 2024 – May 2025*

- Mentored 100+ students during office hours on discrete math, graph theory, and probability distributions; developed standardized benchmarking rubrics for coursework and exams.
- Designed and implemented standardized grading rubrics for exams and assignments, increasing grading consistency and reducing evaluation time by ˜25% while better aligning assessments with learning objectives.
- Delivered targeted, iterative feedback loops that helped 80%+ of students improve performance, strengthening conceptual understanding and reducing common error rates in complex circuit/system analysis.

Projects

AI Interview Evaluation Pipeline | *Python, Mistral, Multi-Agent Systems, JSON Parsing*

- Evaluated 50+ interview transcripts (∼400+ candidate responses) by converting raw conversations into structured JSON cases, scoring answers with Mistral against a rubric, and generating detailed evaluation reports.
- Designed a two-agent loop where Agent 1 scores and Agent 2 audits bias and structure, correcting ∼15–20% of overly harsh ratings and confirming valid evaluator JSON in ∼90% of runs.
- Migrated from a Gemini browser workflow to a Mistral JSON backend, achieving ∼50–60x higher evaluation throughput (∼120–150 cases/min) with robust error handling.

AI Tutor Evaluation Engine | *Python, Redis, MongoDB Atlas, OpenTelemetry, Arize*

- Evaluated ∼1000 AI tutoring sessions with rubric-based scoring on engagement, scaffolding, and goal alignment, producing automated QA feedback reports highlighting tutor strengths and gaps.
- Integrated Redis-backed semantic memory and tracing so tutor sessions are stateful and observable, reducing time to diagnose tutor behavior issues from hours to minutes.
- Built a multi-model verdict engine where multiple LLMs independently score sessions, logging ∼10–20% disagreement cases (∼300+) in MongoDB Atlas as hard examples for model optimization.

Technical Skills

Languages: Python, Java, C++, SQL, JavaScript, TypeScript, Rust, HTML/CSS
AI & Infrastructure: RAG Pipelines, Vector Embeddings, MCP, LLM Agents, Redis, MongoDB Atlas, Docker, AWS, OpenTelemetry, Arize
Frameworks & Tools: React, Node.js, Spring Boot, FastAPI, Flask, PyTest, JUnit, gRPC, Git, Linux